

HW 0

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Chapter 1

1. Markdown, Jinja2, Makefile, SASS, CSS, HTML.

2.

```
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello ,␣world!");
    }
}
```

3. `#include <stdio.h>`

```
int main() {
    printf("Hello ,␣world!");
}
```

Chapter 3

1. One potential edge case that wasn't defined in the book is nested classes. For example, if you were to make a Linked List class, the Node class may be an inner class that doesn't need to exist outside of the Linked List class scope.

Attempting to use inner classes actually leads to a compile error:

```
class LinkedListStack {
    class Node {
        init(data, next) {
            self.data = data;
            self.next = next;
        }
    }

    init() {
        self.head = nil;
    }
    // ...
}
```

[line 2] Error at 'class ': Expect method name.

The correct way to implement this is to write:

```

class Node {
  init(data, next) {
    this.data = data;
    this.next = next;
  }
}

class LinkedListStack {

  init() {
    this.head = nil;
  }

  push(data) {
    if (this.head == nil) {
      this.head = Node(data, nil);
    }
    else {
      var next = Node(data, this.head);
      this.head = next;
    }
  }

  pop() {
    if (this.head == nil) {
      return nil;
    }
    else {
      var out = this.head;
      this.head = this.head.next;
      return out;
    }
  }
}

```

This behaves as I expect since Lox is meant to be a simple language. Nested classes introduce a lot of complexity into the implementation.

2. One unanswered question I have relates to the idea of “truthy-ness”. In languages like Python and JavaScript, certain can evaluate to values when used as the predicate for an “if” statement. Also, these two languages even have different implementations of truthy-ness!

Example in JS:

```

const truthy = []
const falsy = null;

if (truthy) console.log("[] is true!");
if (!falsy) console.log("null is false!");

// [] is true!
// null is false!

```

Example in Python:

```

empty_arr = []
arr = [1, 2, 3]

```

```
if not empty_arr:  
    print("empty_arr_is_false!")
```

```
if arr:  
    print("arr_is_true!")
```

```
# empty_arr is true!  
# arr is true!
```

3. The most important feature that is missing in my opinion is the ability to define return types for functions. I am a big fan of type safety, but I am aware that type safety is quite difficult to implement